Ontology based multi-agent system for financial systemic risk management

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Abstract

Systemic risk refers to the risk or probability of breakdown (losses) in individual parts of components and is evidenced by co-movements (correlation) among most or all parts [1]. The typical case is the subprime mortgage crisis which began in Feb 2007 and hasn't finished yet, brings big loss to nearly all the financial institutions around the world. In this paper, ontology for systemic risk management in financial institutions is proposed and then an ontology based multi agent system is designed to support decision making for systemic risk management in financial institutions.

Keywords
Systemic Risk Management, Ontology, Agent, Multi-agent System

1. Introduction

The event people cares most in financial world during recent two years is American subprime mortgage crisis, which is ongoing economic problem characterized by contracted liquidity in the global credit markets and banking system. Major Banks and other financial institutions around the world have reported losses of approximately U.S. $435 billion as of July 17, 2008 [2] . Subprime mortgage refers to the practice of making loans to borrowers who do not qualify for market interest rates owing to various risk factors, such as income level, size of the down payment made, credit history, and employment status. Why subprime mortgage in U.S causes such a big loss to the world? This is because of systemic risk in financial world. Systemic risk refers to the risk or probability of breakdown (losses) in individual parts of components and is evidenced by co-movements (correlation) among most or all parts [1, 3].

In this study, an ontology based multi agent system is proposed for systemic risk management in financial institutions. With it, systemic risk signal can be detected from news gathering and analysis.

Our work consists of the natural integration of two fields of computing techniques. The field of ontology provides our research with a formal framework with which knowledge regarding the problem domain can be specified in a concise and unambiguous manner [4]. On the other hand, the recent flurry of research into the application of intelligent agents has resulted in providing distributed, intelligent agents capable of dealing with vast amounts of information collaboratively.

The rest of the paper is organized as follows: Section 2 discusses the related technology used in this research; the ontology for systemic risk is proposed and discussed in Section 3; and then an ontology based system is designed and implemented in Section 4; finally all is concluded in Section 5

2. Related Techniques

2.1 Ontology

Ontology is a branch of philosophy concerned with the study of what exists. In artificial intelligence, ontologies can be used to encourage standardization of the terms to represent knowledge about a domain [5]. And when ontologies are formalized in first-order logic or a subset thereof, they can support inference mechanisms[6]. An OWL (Web Ontology Language) is designed for satisfying the need to process the content of information instead of just presenting

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information[1]. It allows more vocabulary terms to be used for describing properties and classes.

Subclass of: it links subclass to its parent class.

Object property: it represents the relationship among classes.

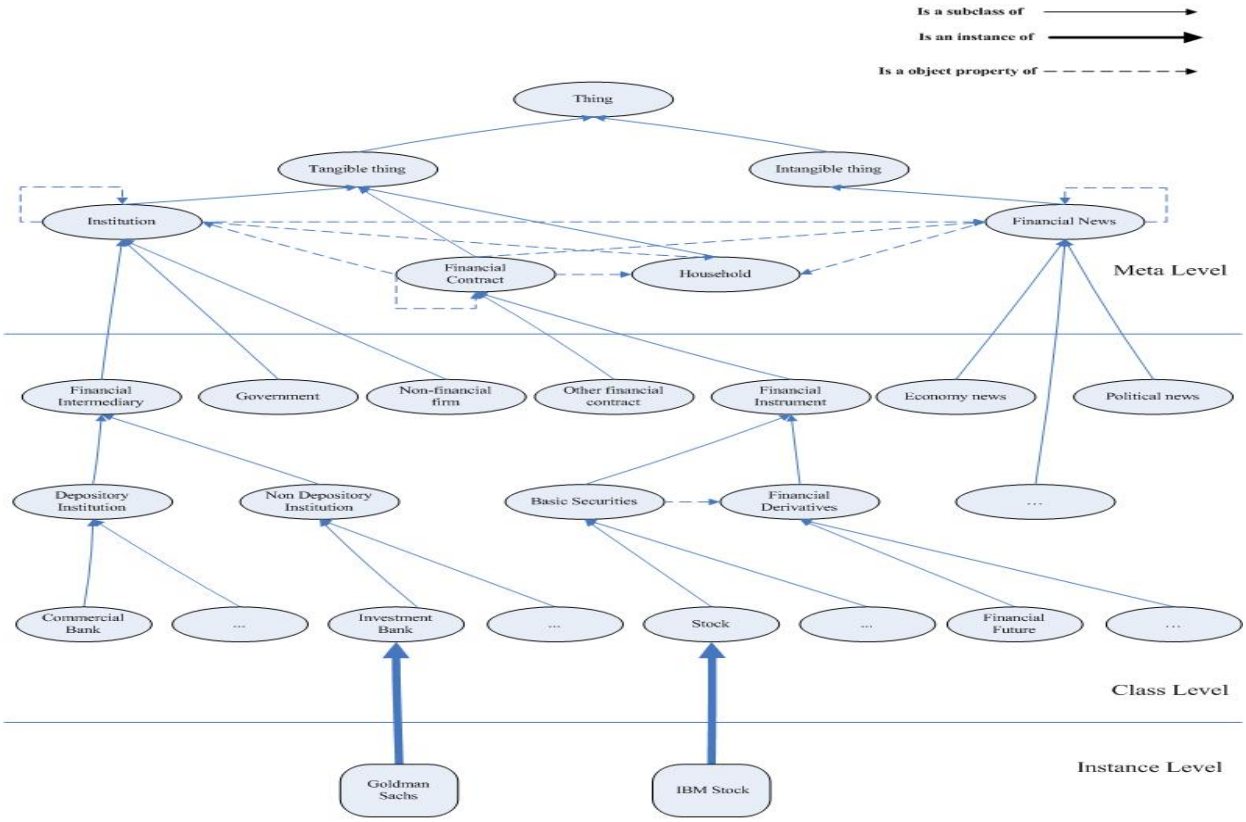

Figure 1. Ontology for systemic risk management in financial institutions

2.2 Intelligent agent-assisted decision support system

An agent is defined as a computer system that is situated in some environment, and is capable of autonomous action in that environment in order to meet its design objectives [7] . In recent years, there has been considerable growth of interest in the design of a distributed, intelligent society of agents capable of dealing with complex problems and vast amounts of information collaboratively [8, 9]. The potential contributions of intelligent agents to decision support systems (DSSs) have been described as enormous [10].

3. Ontology for financial systemic risk management

Figure 1 shows the concepts of ontology for financial systemic risk management. Ellipse represents the concept in ontology and three different arrow means three kinds of relationships between concepts.

Instance of: it links instance to the class belonging and then inherit its every property.

The ontology was further discussed through three different levels: meta level, class level and instance level.

Meta level

Four key concepts are shown in the meta level which are institution, financial, household and financial news.

The arrows in the meta level shows that these four concepts have complex relationships which denoted by object properties. And for institution, financial contract and financial news, there are also relationships between their subclasses. However, the detail of relationship will be discussed in class level.

Class level

Class level gives the detail classification of concepts. Take institution for example, it has been proposed that the basic institutions in financial market are non-financial firms which produce normal goods and service, financial intermediary who earn money though providing financial service and government which also need financial market to issue securities

[1] http://www.w3c.org/2004/owl/

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[11]. Furthermore, financial intermediary includes depository institutions (such as commercial bank, mutual saving banks, credit unions …) and non depository institution (such as investment bank, insurance companies, mutual funds, ..) [12].

The concepts in class level inherit the properties in meta level, and the detail of these properties will be discussed.

Instance level

Some instances are given in this level, for example Goldman Sachs is an instance of investment bank, and IBM stock is an instance of stock.

developing criteria, and identifying the various alternatives through which the problem can be solved. In the choice phrase, the decision maker chooses the best alternative that meets the criteria, and makes the final decision. Following these three phrased, the decision maker uses the feedback from the results of the decision to review how well the process is executed. Such reflection on past processes can form a basis of the intelligence phase for future decisions. In this study, Simon's decision-making process theory is employed in designing architecture of FSRMS, showing in Fig. 2.

Service-Oriented Architecture (SOA) idea is also adopted in this research, and then agents are classified

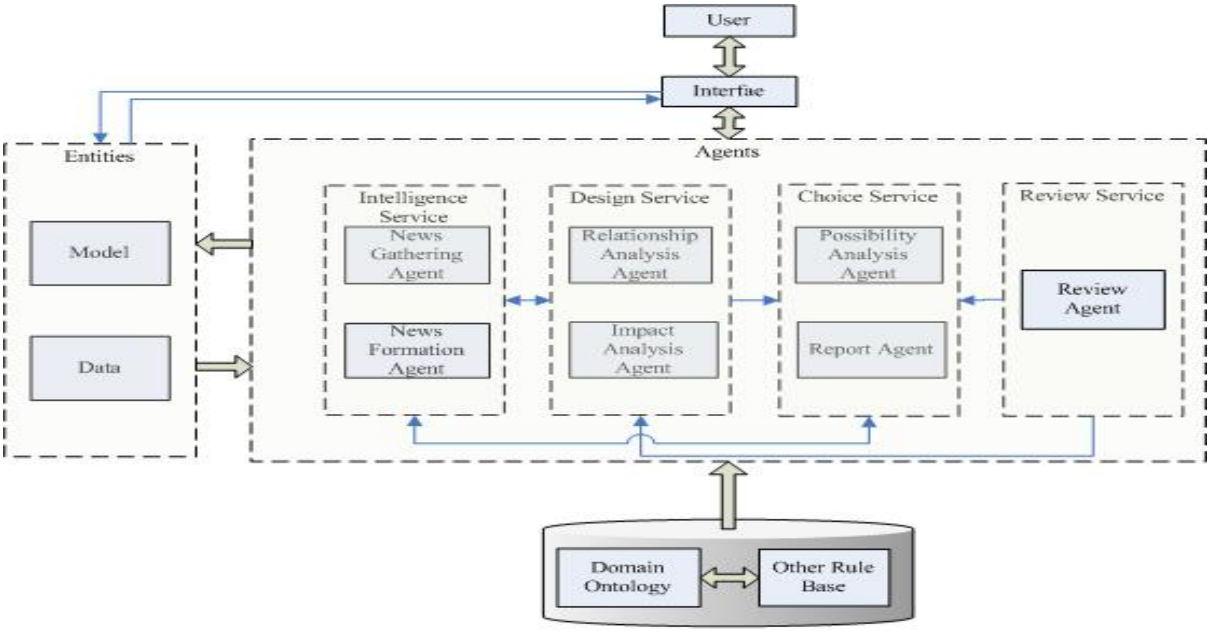

Figure 2. System Architecture diagram of financial systemic risk management system

4. Ontology based multi-agent system for financial systemic risk management

This section presents the design and development of a novel prototype of multi agent financial systemic risk management design support system (FSRMS) on the basis of the proposed ontology.

4.1 system design architecture

Our proposed intelligent FSRMS provides real time systemic risk report based on gathered financial news analysis. In his classic work, Herbert A. Simon proposed a decision process comprising four distinct phases – intelligence, design, choice and review [13]. In the intelligence phase, the decision maker recognized the problem at hand and gathers information about the situation. The design phase is marked by structuring the problematic situation,

into four different groups providing Intelligent Service, Design Service, Choice Service and Review Service [14]. Intelligent Service is provided by two agents that obtain, aggregate, and assess relevant financial news. All the Design, Choice and Review Service may request information relating to their task from the Intelligence Service, if required. The Design Service is done by Relationship Agent and Impact Analysis Agent. With them, financial news is analyzed whether it is related to the applied financial institution. And if yes, how much is the impact? In order to keep simple, four weights respectively denote impact, bid impact, litter impact and unclear. Then Choice Service is provided by Possibility Analysis Agent and Report Agent. With them, system infers the possible systemic risk event and reports the result. Finally, Review Service is used to get feedback from every reporting process and it makes the system have learning ability.

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The ontology discussed in section 3 is used as part of knowledge base in designed FSRMS. For example, financial news ontology can be used to support news gathering and formatting agents, as it provides a formal conceptualization of financial news. Financial institution ontology can be used to analyze which institution is related to gathered financial news.

4.2 System implementation

At the bottom level, the operating system is Linux. JESS (Java Expert System Shell and Scripting language) rules are adopted as basic implementation language [15]. In the prototype system, each agent contains a JESS rule set for reasoning. The reasoning results are asserted JESS facts. A popular language for agent communication, Knowledge query and Manipulation, is also employed.

Further more, Protégé, which is an open platform for ontology modeling and knowledge acquisition[2], is used to edit ontology discussed in above section into JESS. SWRLJessTab is a Protégé plug-in intended to bridge between Protégé OWL, RACER and Jess, for reasoning with both OWL and SWRL [16]. Using SWRLJessTab, it is possible to infer knowledge from an OWL ontology combined with a SWRL rule base, both represented in Protégé OWL. So ontology proposed in section 3 can be first edited in protégé with OWL format and then reasoned in JESS environment.

5. Conclusion

The ongoing subprime mortgage crisis indicates the importance of systemic risk management in financial institutions. In this paper, ontology for financial systemic risk management is proposed and then an ontology based multi-agent system is designed and implemented. The contributions of this research are as following:

. From application perspective, a decision support system is proposed. With it, the financial institution can get systemic risk report based on financial news analysis.

. From technology perspective, the two technology ontology and multi agent system are integrated together in a real world project, which use OWL, JESS and Protégé to implement them into a final decision support system. So the whole methodology will also benefit system development in any other domain.

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